

CSL7870: TASK-09 — Binary Classification on AIDS Using Graph2Vec (WL vs Anonymous Walk Documents)

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Abstract

I study graph-level embeddings for the **AIDS antiviral screen** dataset. Each molecular graph is labeled *active* (class 1) or *inactive* (class 0) against HIV. I first remove structural duplicates via a WL-based isomorphism signature (Task-7 choice: betweenness-seeded). I then build graph “documents” in two ways: (i) WL-kernel refinements and (ii) Anonymous Walks (AW). I train Doc2Vec embeddings and evaluate a shallow classifier on a held-out test split. I complete all required sweeps: WL dimension ($16 \rightarrow 1024$), AW walk length L ($5 \rightarrow 105$), AW number of walks M ($10 \rightarrow 1010$), and a coarse ($L \times M$) grid. I report macro and class-wise metrics, explain the jagged precision seen in AW #walk sweeps, and identify the best AW configuration. WL scales smoothly with dimension; AW benefits strongly from longer walks. Best WL (at $d=1024$) achieves macro-F1 **0.9441**. Best AW from the one-dimensional sweeps (at $L=105, M=100$) reaches macro-F1 **0.9520**; the coarse grid’s best ($L=105, M=410$) gives macro-F1 **0.9453**. I discuss when each representation is preferable.

1 Dataset and Pre-processing

- **Dataset.** AIDS antiviral screen (graphs are molecules; nodes: atoms; edges: bonds). Label: **class 1** (active), **class 0** (inactive).
- **Isomorphism removal.** I compute a 3-iteration WL signature with node labels set to betweenness centrality and keep one representative per signature.
- **Class balance after de-duplication.** Class 0: **320**; Class 1: **858** (see Fig. 1).

2 Methods

2.1 Graph Documents

WL documents.

- I seed node labels with the **degree** when constructing WL documents for Doc2Vec (simple, stable vocabulary growth).
- I then apply WL refinement for a fixed number of iterations.

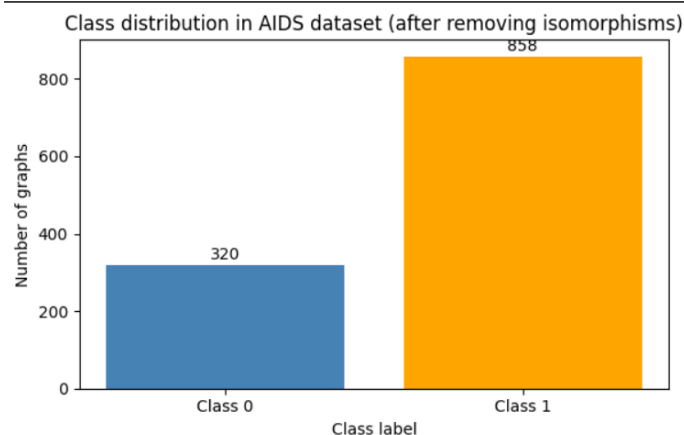


Figure 1: Class distribution after removing isomorphic copies (class 0: inactive; class 1: active).

- Each iteration yields a sentence of node labels; concatenating sentences forms the document.
- *Note:* For **isomorphism removal** (Task-7 choice) I seed with **betweenness centrality** before WL refinement; that step is only for de-duplication.

Anonymous Walk (AW) documents.

- For each graph, I sample M random walks of length L (a walk with L edges has $L+1$ tokens).
- I use a *visit-count* anonymization: the token at step t equals the number of times the current node has appeared up to t in that walk.
- Each walk is a sentence; all walks together form the document.
- *Note:* This differs from the more common *first-occurrence* AW (0,1,0,2,...). I keep visit-count AW for this task and leave a direct comparison for future work.

2.2 Embeddings and Classifier

- **Doc2Vec** (PV-DM) learns a d -dimensional graph vector. For WL I sweep $d \in \{16, 32, \dots, 1024\}$. For AW I use $d=128$ in the one-dimensional sweeps and in the coarse grid.
- **Classifier.** A small MLP trained on frozen embeddings. I use an 80/20 stratified split and report all metrics on the **test** subset.

2.3 Metrics and Interpretation

Let TP/FP/FN/TN denote true/false positives/negatives for *class 1*.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

- **Precision:** “Of compounds predicted active, how many are truly active?” (low FP)
- **Recall:** “Of truly active compounds, how many did I find?” (low FN)
- **Macro** averages give equal weight to both classes and are robust to imbalance.

3 Experiments

1. **WL baseline (128-dim)**. Report Accuracy, Precision, Recall, F1.
2. **WL dimension sweep**. $d \in \{16, 32, \dots, 1024\}$.
3. **AW baseline (128-dim)**. Report metrics for a chosen (L, M) .
4. **AW length sweep**. $L \in \{5, 15, \dots, 105\}$ at $M=100$.
5. **AW #walks sweep**. $M \in \{10, 110, \dots, 1010\}$ at $L=5$.
6. **Coarse grid (extension)**. $(L \times M)$ with $L \in \{5, 25, 45, 65, 85, 105\}$, $M \in \{10, 210, 410, 610, 810, 1010\}$.

4 Results and Analysis

4.1 WL: Baseline and Dimension Sweep

WL@128 (test).

- Accuracy: **0.9364**
- Precision (macro): **0.9599**
- Recall (macro): **0.8828**
- F1 (macro): **0.9127**

Dimension trend (test).

- Metrics generally increase with d ; best WL occurs at $d=1024$ with:
 - Accuracy **0.9576**, Precision (macro) **0.9661**, Recall (macro) **0.9268**, F1 (macro) **0.9441** (*see Fig. 2*).
- Small non-monotonicity around mid-dims is expected on a single split with Doc2Vec stochasticity.

4.2 AW: Walk Length Sweep L (at $M=100$, $d=128$)

Observation (test).

- All macro metrics improve as L increases (*see Fig. 3*).
- The best point of the sweep is $L=105$ with:
 - Accuracy **0.9619**, Precision (macro) **0.9498**, Recall (macro) **0.9542**, F1 (macro) **0.9520**
- Precision peaks slightly earlier (near $L=95$), indicating recall gains dominate at the longest walks.

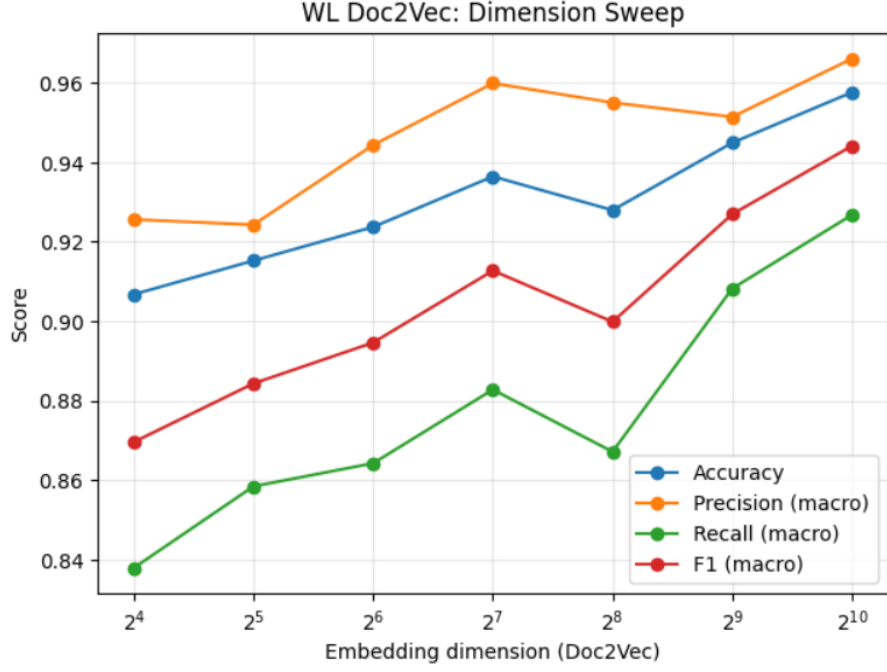


Figure 2: WL Doc2Vec dimension sweep ($16 \rightarrow 1024$) (test set).

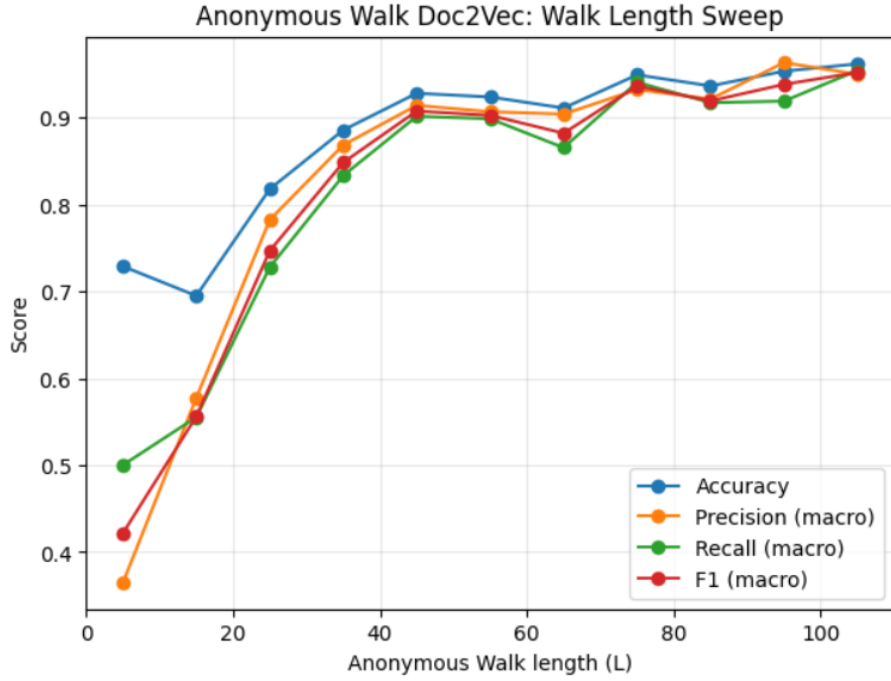


Figure 3: AW Doc2Vec: sweep over walk length L at $M=100$, $d=128$ (test set).

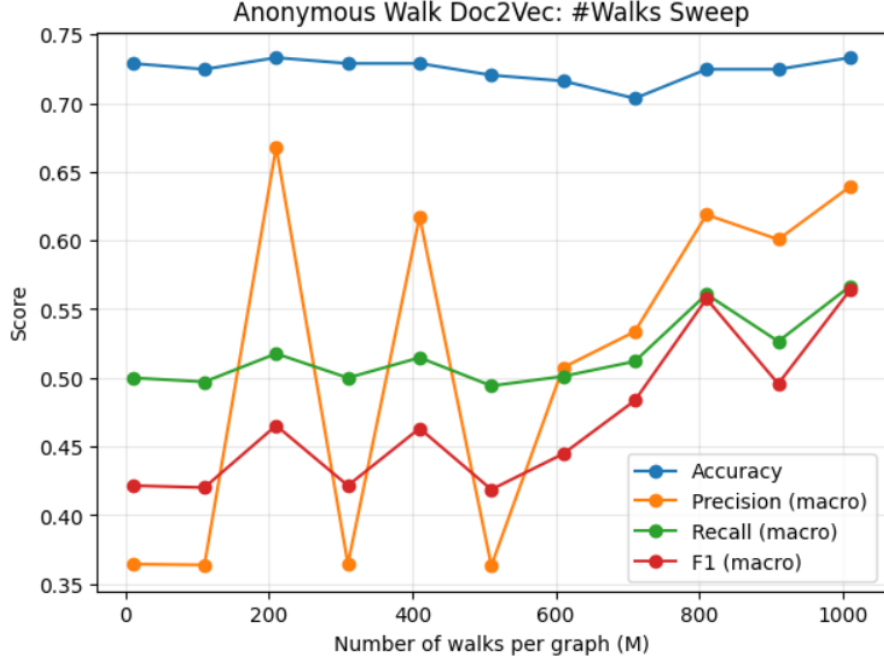


Figure 4: AW Doc2Vec: sweep over number of walks M at $L=5$, $d=128$ (test set).

4.3 AW: Number of Walks Sweep M (at $L=5$, $d=128$)

Observation (test).

- Accuracy/Recall/F1 vary moderately with M , but macro **precision is jagged** (see Fig. 4).
- Class-wise analysis later shows the jaggedness originates from **class 0** (inactive); class 1 remains high and stable.

4.4 WL(best) vs AW(best) from 1-D sweeps

Selection rule.

- WL(best) = best WL across d ($d=1024$, macro-F1 **0.9441**).
- AW(best-1D) = best AW across the two one-dimensional sweeps (here it comes from the L -sweep at $L=105$, $M=100$, macro-F1 **0.9520**).

Finding (test).

- AW(best-1D) outperforms WL(best) on Accuracy, Recall, and F1; WL retains a slight edge on macro precision in some settings (see Fig. 5).

4.5 Class-wise Behaviour (Diagnosing Precision Jaggedness)

Per-class at AW $L=105$, $M=100$ (test).

- Class 0: Precision **0.9231**, Recall **0.9375**, F1 **0.9302**

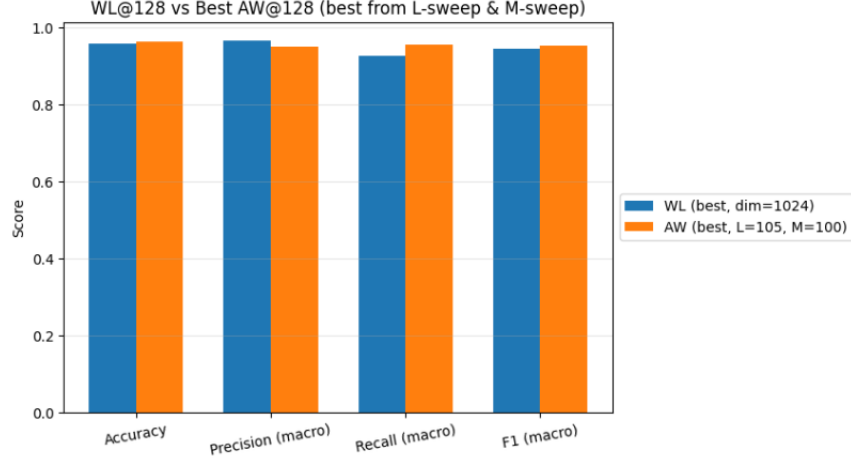


Figure 5: Head-to-head: WL(best) vs AW(best-1D) at $d=128$ for AW and best d for WL (test set).

- Class 1: Precision **0.9766**, Recall **0.9709**, F1 **0.9738**
- Confusion: TN **60**, FP **4**, FN **5**, TP **167** (see Fig. 6 and Fig. 7).

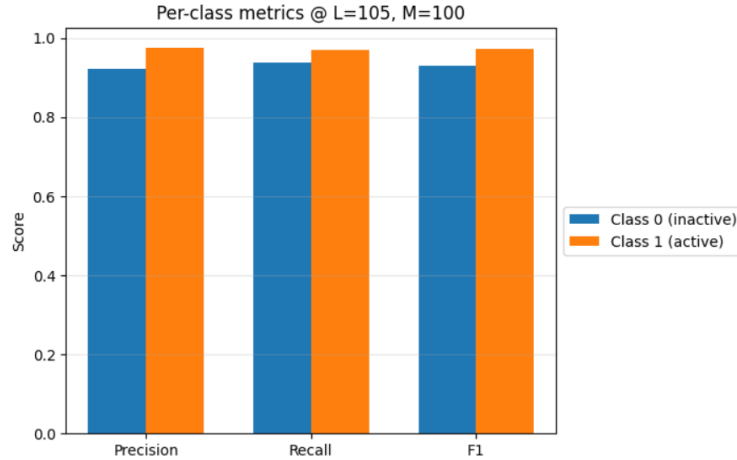


Figure 6: Per-class metrics for AW at $L=105, M=100$ (test set).

4.6 Class-wise Sweeps (What the macro curves hide)

WL: per-class vs dimension.

- Both classes improve with d ; the class gap (Class 1 minus Class 0) narrows slightly at higher d (see Fig. 8).

AW: per-class vs L .

- Class 0 improves steadily with longer walks; disparity $\Delta F1$ reduces as walks capture larger motifs (see Fig. 9).

AW: per-class vs M .

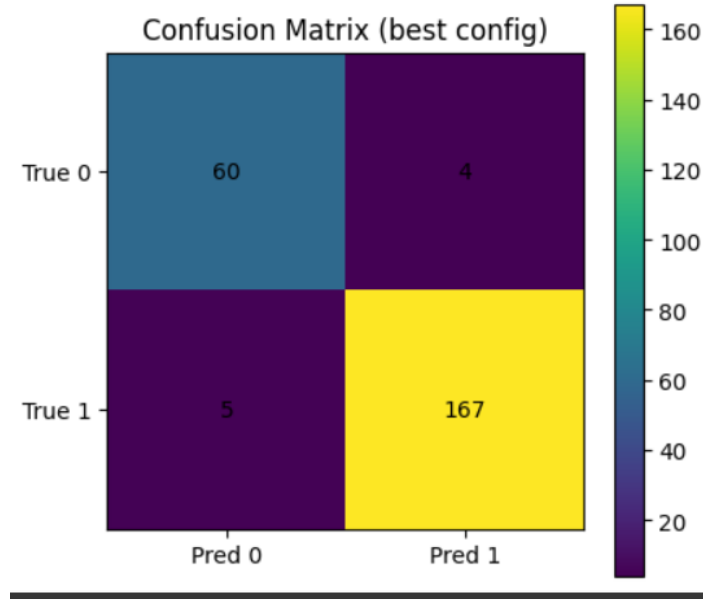


Figure 7: Confusion matrix for AW at $L=105, M=100$ (test set). Few FP/FN; class 0 support is smaller.

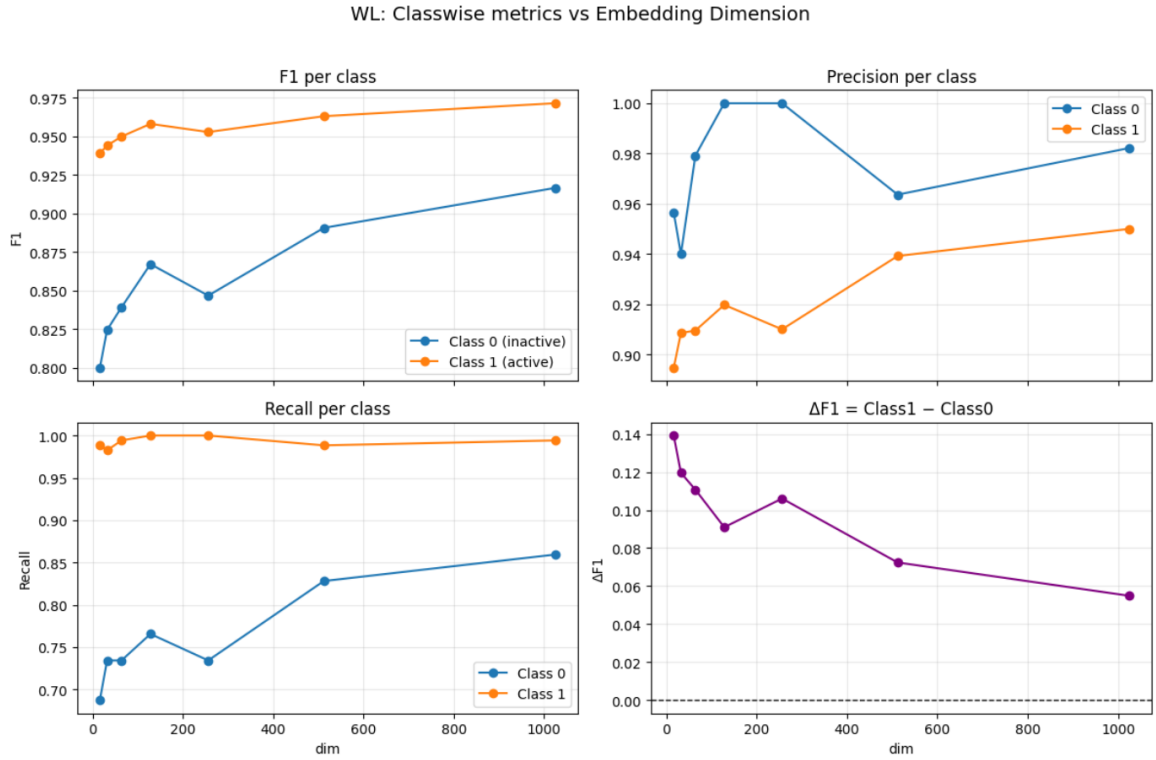


Figure 8: WL class-wise metrics vs embedding dimension (test set).

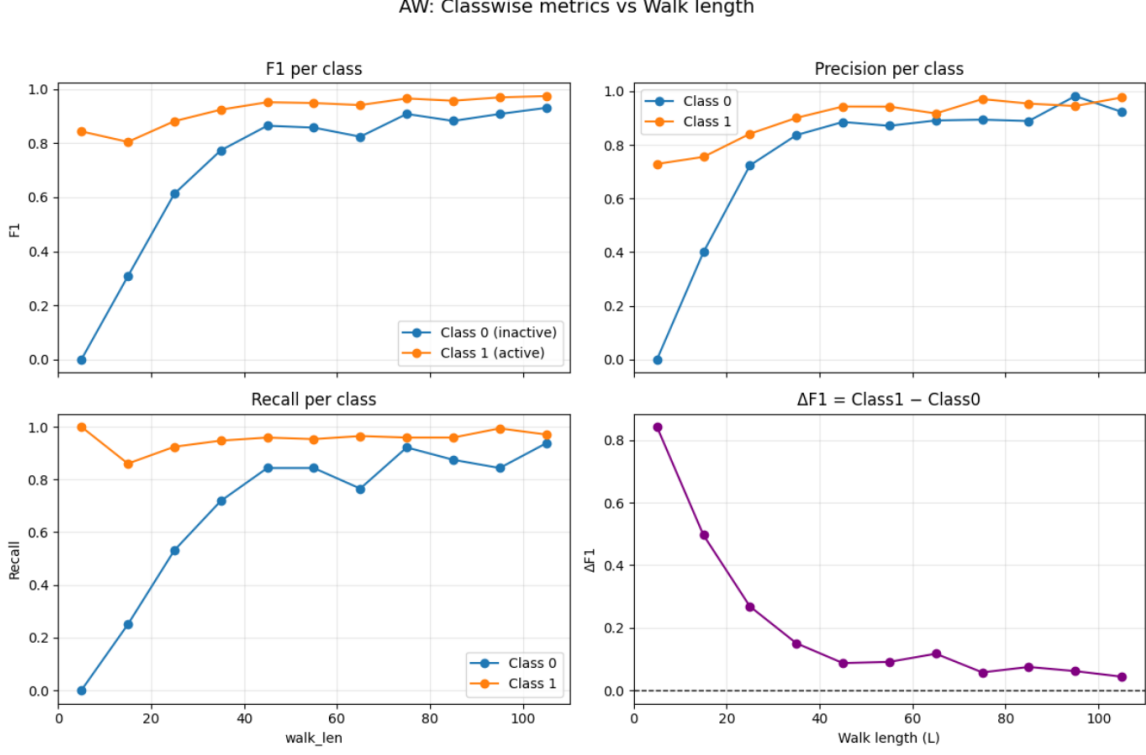


Figure 9: AW class-wise metrics vs walk length L (test set).

- Class 0 precision oscillates sharply across M ; class 1 stays high and stable. This explains the jagged macro precision in Fig. 4 and aligns with Fig. 10.

4.7 Anonymous Walk ($L \times M$) Grid (Coarse)

Grid and winner (test).

- Grid: $L \in \{5, 25, 45, 65, 85, 105\}$, $M \in \{10, 210, 410, 610, 810, 1010\}$, $d=128$.
- **Best coarse AW:** $L=105, M=410$ with:
 - Accuracy **0.9576**, Precision (macro) **0.9551**, Recall (macro) **0.9366**, F1 (macro) **0.9453**
- Naïve comparison across all examined settings:
 - WL(best over d): F1 **0.9441** at $d=1024$
 - AW(best-1D): F1 **0.9520** at $(L=105, M=100)$
 - AW(best-grid): F1 **0.9453** at $(L=105, M=410)$
- **Final winner among examined points:** AW(best-1D) at $(L=105, M=100)$ with F1 **0.9520**. The coarse grid’s winner is very close and favors slightly higher precision with a small F1 drop.

4.8 Runtime vs Quality (Cost View)

I summarize F1 vs wall-clock time (the `sec` column) across sweeps:

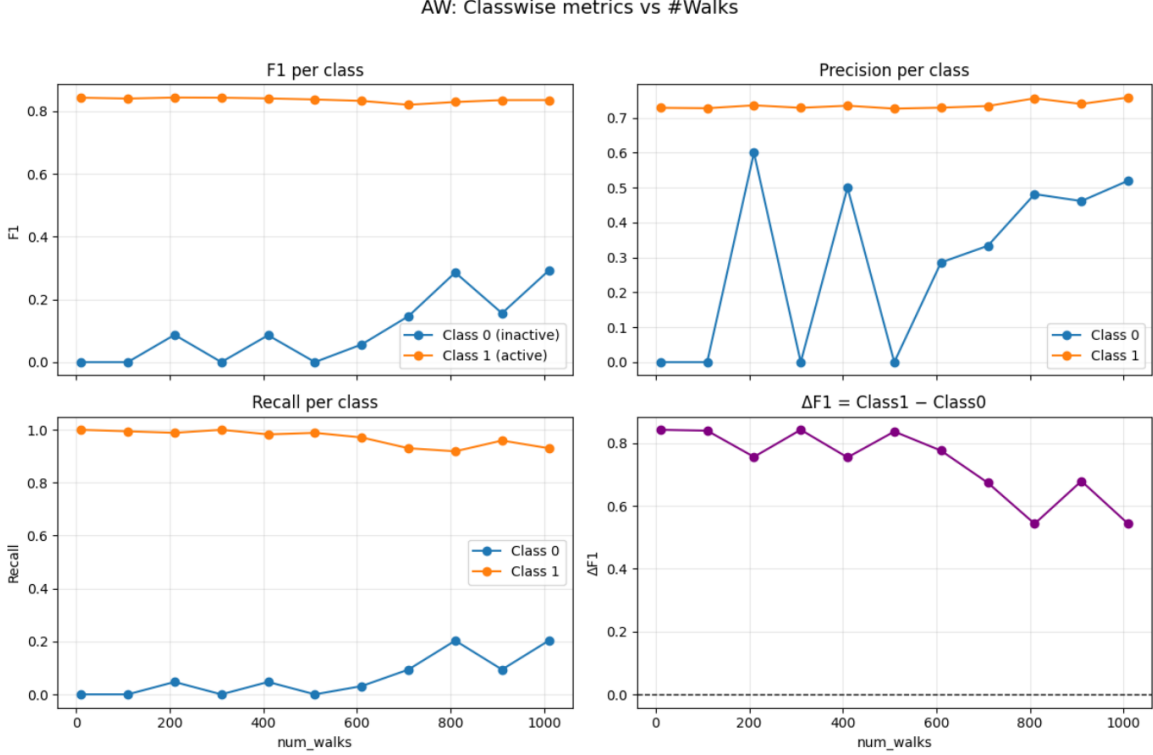


Figure 10: AW class-wise metrics vs number of walks M (test set).

- **WL (dims 16→1024).** Runtime stays tight ($\sim 2.95\text{--}3.95\text{ s}$); macro-F1 rises from **0.8696** (@16) to **0.9441** (@1024).
- **AW: increase L (fix $M=100$).** Runtime grows roughly linearly: **5.42 s** (@ $L=5$) $\rightarrow$ **60.49 s** (@ $L=105$), while macro-F1 jumps **0.42** $\rightarrow$ **0.95**. L is the efficient lever.
- **AW: increase M (fix $L=5$).** Runtime **3.34 s** (@ $M=10$) $\rightarrow$ **26.68 s** (@ $M=1010$), but macro-F1 only **0.42** $\rightarrow$ **0.56**. M is far less impactful at short walks.
- **Takeaway.** If I can afford more compute, spending it on *longer walks* gives large, reliable gains; simply adding more short walks has poor returns.

5 Discussion

- **Mechanism: WL vs AW (why they behave differently).**
 - **WL** aggregates locally consistent multisets; with few iterations it stresses near-neighborhood agreement, which naturally **preserves precision** (fewer borderline positives).
 - **AW** with larger L encodes broader motifs (rings, fused rings, functional-group contexts). This expands positive coverage and **boosts recall** strongly—exactly what I observe as L increases (macro-Recall $\uparrow$; Fig. 3 and Fig. 9).
- **Why macro precision looks jagged across M at $L=5$ (numeric, class-wise).**
 - The test split has only ~ 64 negatives (class 0). At the best AW point ($L=105$, $M=100$), I have **TN=60**, **FP=4**, **FN=5**, **TP=167** (Fig. 7); class 0 precision is $\frac{60}{60+5} = \mathbf{0.923}$.

- A swing of just ± 2 errors shifts class 0 precision substantially: $\frac{60}{60+7}=\mathbf{0.896}$ or $\frac{60}{60+3}=\mathbf{0.952}$. With $L=5$ the features are weak, so small sampling changes from increasing M move the boundary and produce the **macro-precision oscillations** (Fig. 4, Fig. 10).
- **Cost view (F1 vs seconds).**
 - **WL** gives steady gains while staying $\approx 3\text{--}4\text{ s}$ end-to-end (Fig. 2).
 - **AW**, when I *grow* L , goes from **5.4 s** to **60.5 s** but delivers the largest F1 improvements (to **0.952**).
 - **AW**, when I *grow* M at short L , costs up to **26.7 s** with minor F1 gain ($\leq \mathbf{0.564}$). If compute is limited, I should **increase L first**, not M .
- **Tying class-wise curves to design choices (what the macro curves hide).**
 - **WL vs dimension:** increasing d reduces the **Class 1 vs Class 0 gap** (Fig. 8), consistent with higher-capacity embeddings capturing subtler local patterns that help the minority class.
 - **AW vs L :** Class 0 benefits the most from longer walks (Fig. 9); the $\Delta\mathbf{F1}$ **shrinks** as L grows, matching the intuition that larger structural context reduces FN on difficult negatives.
- **When to prefer which?**
 - If I prioritize **finding more actives** (recall/F1), use **AW with long L** .
 - If I prioritize **fewer false actives** (precision) or need **very fast** runs, **WL** is a compelling baseline.

6 Limitations

- Single 80/20 split; seed averaging or cross-validation would reduce variance and smooth curves.
- AW anonymization uses visit-count tokens; canonical first-occurrence anonymization could be evaluated.
- The $(L \times M)$ grid is coarse for runtime; a local refinement around $(105, 410)$ may yield a small gain.

7 Conclusion

- I implemented both required functions, removed isomorphisms, and completed all sweeps with statistical charts plus a coarse 2-D grid.
- WL is a strong, steady baseline; AW becomes competitive and often superior on recall/F1 with longer walks.
- Across examined settings, the top F1 is achieved by AW at $(L=105, M=100)$; the grid’s best $(105, 410)$ is a close second with a precision tilt.

Appendix: Key Tables (test split)

Table 1: WL@128 baseline.

	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
WL@128	0.9364	0.9599	0.8828	0.9127

Table 2: Best configurations and scores among examined points.

Model	Config	Accuracy	Recall (macro)	F1 (macro)
WL(best)	$d=1024$	0.9576	0.9268	0.9441
AW(best-1D)	$L=105, M=100$	0.9619	0.9542	0.9520
AW(best-grid)	$L=105, M=410$	0.9576	0.9366	0.9453